Operating Systems Lab Assignment 4

Problem : Write a C program which will be able to read a file using read() system call. Now you provide a mechanism to set a lock by using fcntl() system call and to release the lock by using the same fcntl() system call. For setting up and releasing out a lock, you have to fill up a structure flock (struct flock) with proper values. Now you run the program from two different terminals simultaneously and demonstrate that if one process has got the lock, then that process only can read the file. But the other one is not permitted to read the file until the lock is released by the first process. That means, only one process who acquires the lock can read the file at any instant of time

Ans:

Testfile.txt : Process communication through file locking.

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/types.h>

int main() {
    int fd;
    char buffer[100];
    struct flock lock;

    fd = open("testfile.txt", O_RDWR);
    if (fd == -1) {
        perror("open");
        exit(1);
    }

    lock.l_type = F_WRLCK;
    lock.l_whence = SEEK_SET;
    lock.l_start = 0;
    lock.l_len = 0;
    lock.l_pid = getpid();

    printf("Process %d: Attempting to get lock...\n", getpid());

    if (fcntl(fd, F_SETLKW, &lock) == -1) {
        perror("fcntl");
        exit(1);
    }

    printf("Process %d: Lock acquired. Reading file...\n", getpid());

    int bytes_read = read(fd, buffer, sizeof(buffer) - 1);
    if (bytes_read > 0) {
        buffer[bytes_read] = '\0';
        printf("Process %d content: %s\n", getpid(), buffer);
    }

    printf("Process %d: Sleeping for 10 seconds...\n", getpid());
    sleep(10);

    lock.l_type = F_UNLCK;
    if (fcntl(fd, F_SETLK, &lock) == -1) {
        perror("fcntl unlock");
        exit(1);
    }

    printf("Process %d: Lock released.\n", getpid());
    close(fd);

    return 0;
}
```

Output:

```
Process 3: Attempting to get lock...
Process 3: Lock acquired. Reading file...
Process 3 content: Process communication through file locking.
Process 3: Sleeping for 10 seconds...
```